Mengzhen Li

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Education

University of Minnesota, Minneapolis, MN Aug. 2021 – Expected May 2027
M.S./Ph.D. Computer Science, GPA: 3.71/4.0

College of Saint Benedict, Saint Joseph, MN Jan. 2018 – May 2021
B.A. Computer Science, Summa Cum Laude, GPA: 3.93/4.0

Experience

Research Lead, Medtronic-Sponsored Industry Collaboration June 2025 – June 2026
Medtronic and University of Minnesota *Minneapolis, MN*

- Led a **12-month, industry-funded** research project with Medtronic, a global medical-technology leader, from problem definition through delivery, designing **automated log analysis** and **sequence-based anomaly detection** to accelerate failure management and root-cause analysis.
- Built an **LSTM-based** anomaly detector over **15M+ real-world records;** caught **105 confirmed bugs** in operational data.
- **System adopted by Medtronic engineers for internal use**, supporting faster failure triage and root-cause analysis for **safety-critical medical technology**.

Graduate Research Assistant (Ph.D. Candidate) Aug. 2021 – Present
University of Minnesota *Minneapolis, MN*

- Research focus: **automated repair and maintenance of software test code**. Built three tools that find and fix defects in Java test suites without changing program behavior, validated on **500+ real-world open-source projects**.
- **ARUS**: Test code often contains unused mock setups (unnecessary stubbings) that hide bugs and slow down maintenance. Built a tool that analyzes runtime executions to **safely identify and remove them**, generating behavior-preserving fixes across **128 open-source projects**.
- **REMI**: Detects and fixes **mismatched method invocations** in tests, where a test does not check what it claims to. Evaluated on **217 projects**; automatically fixed **171 of 172 issues (99%)** with minimal code changes; **7 patches merged** by maintainers.
- **REDUST**: Test suites accumulate heavy copy-pasted setup code. Built a tool that refactors it automatically, consolidating **204 duplicated configurations into 68 reusable helpers (67% fewer)** across **213 projects** while preserving behavior.
- **Real-world impact: 85 pull requests merged** into open-source projects including Google-oauth (Jenkins) and LDBC, accepted by maintainers as production changes.

Publications and Awards

[Distinguished Paper Award] **Mengzhen Li**, Mattia Fazzini. "Automatically Removing Unnecessary Stubbings from Test Suites." *IEEE ICST 2024.*

Mengzhen Li, Mattia Fazzini. "Automatically Resolving Mismatched Invocations on Test Doubles in Test Suites." *IEEE ICST 2026.*

Jamie Lee, Flynn Teh, Hengcheng Zhu, **Mengzhen Li**, Mattia Fazzini, Valerio Terragni. "Improving LLM-Driven Test Generation by Learning from Mocking Information." *IEEE ICSTW 2026.*

Mengzhen Li, Mattia Fazzini. "DataTD: A Dataset of Java Projects Including Test Doubles." *IEEE/ACM MSR 2025.*

Alessio Gambi, **Mengzhen Li**, Mattia Fazzini. "Artisan: An Action-Based Test Carving Tool for Android Apps." *IEEE ICSME 2023.*

Technical Skills

Languages: Java, Python, C++, SQL, JavaScript, SML
Tools and Platforms: Git, Docker, Google Cloud Platform, Linux